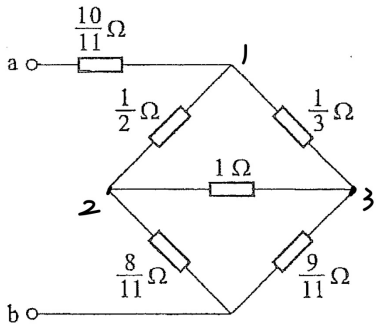
2-6 利用 Y-Δ 等效变换求题 2-6 图中 a、b 端的等效电阻。

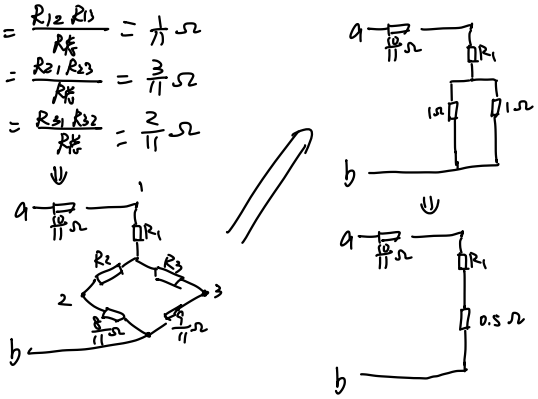


题 2-6 图

$R_1 = \frac{R_{12}R_{13}}{R_{\text{总}}} = \frac{1}{11}\Omega$

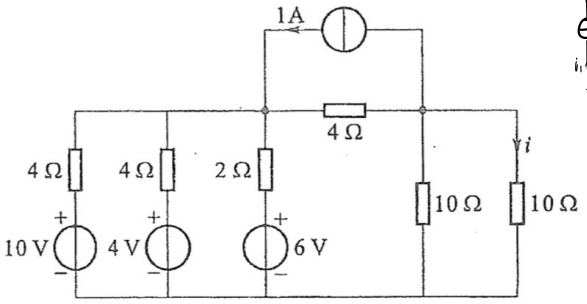
$R_2 = \frac{R_{21}R_{23}}{R_{\text{总}}} = \frac{3}{11}\Omega$

$R_3 = \frac{R_{31}R_{32}}{R_{\text{总}}} = \frac{2}{11}\Omega$

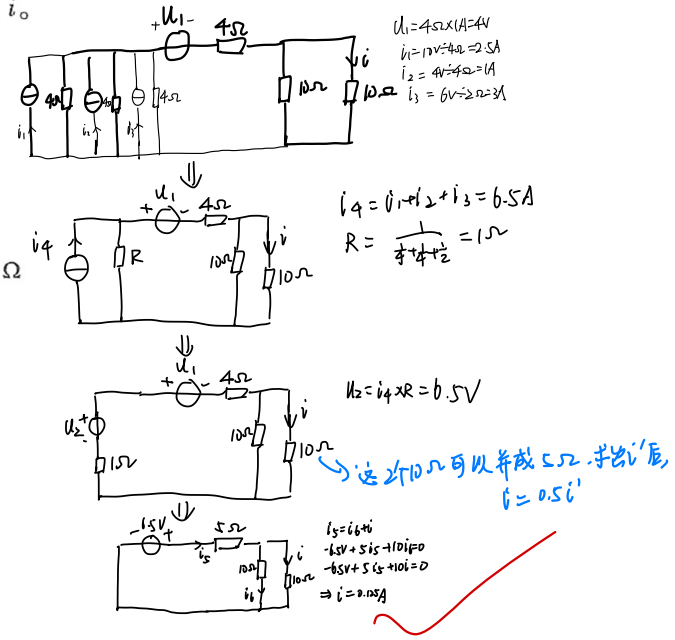


$\Rightarrow R_{ab} = \frac{10}{11}\Omega + \frac{1}{11}\Omega + 0.5\Omega = 1.5\Omega$

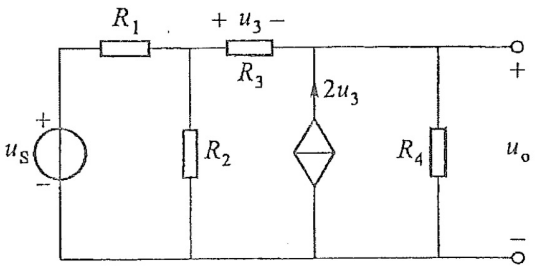
2-11 利用电源的等效变换，求题 2-11 图所示电路的电流 i。



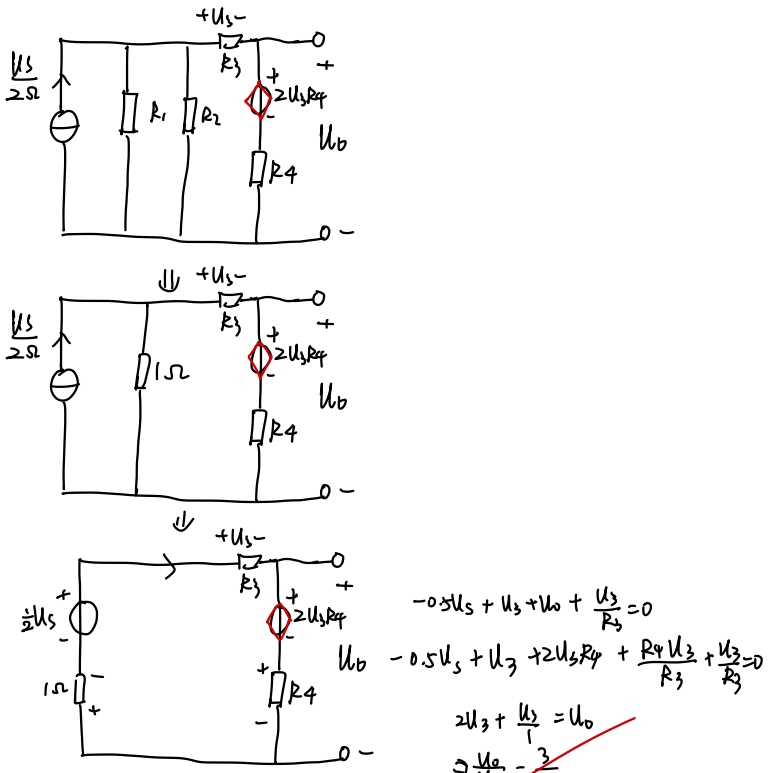
题 2-11 图



2-12 利用电源的等效变换，求题 2-12 图所示电路中的电压比 $\frac{u_o}{u_S}$。已知 $R_1 = R_2 = 2\ \Omega$，$R_3 = R_4 = 1\ \Omega$。



题 2-12 图



$-0.5U_S + U_3 + U_o + \frac{U_3}{R_3} = 0$

$U_o \quad -0.5U_S + U_3 + 2U_3R_4 + \frac{R_4U_3}{R_3} + \frac{U_3}{R_3} = 0$

$2U_3 + \frac{U_3}{1} = U_o$

$\Rightarrow \frac{U_o}{U_S} = \frac{3}{10}$

2-15 试求题 2-15 图的输入电阻 R_{ab}。

(a) $U_{ab} + R_2 i - \mu u_1 + u_1 = 0$

$\Rightarrow \frac{U_{ab}}{i} + R_2 - \mu \frac{u_1}{i} + \frac{u_1}{i} = 0$

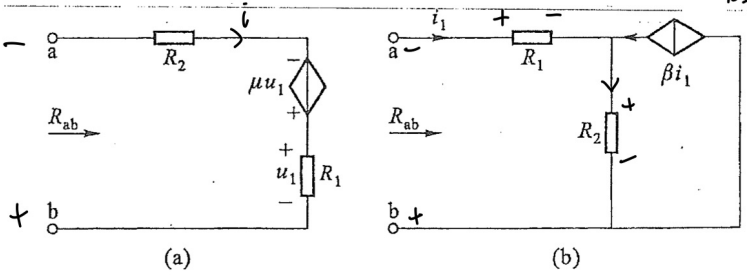
$R_{ab} + R_2 - \mu R_1 + R_1 = 0$

$R_{ab} = \mu R_1 - R_1 - R_2$

(b) $U_{ab} + i_1 R_1 + (i_1 + \beta i_1) R_2 = 0$

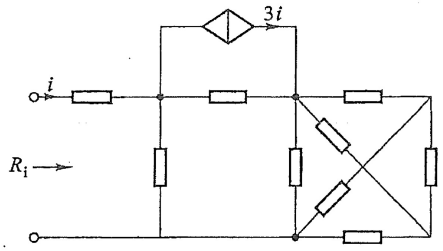
$\frac{U_{ab}}{i_1} + R_1 + (1+\beta) R_2 = 0$

$R_{ab} = -R_1 - (1+\beta) R_2$

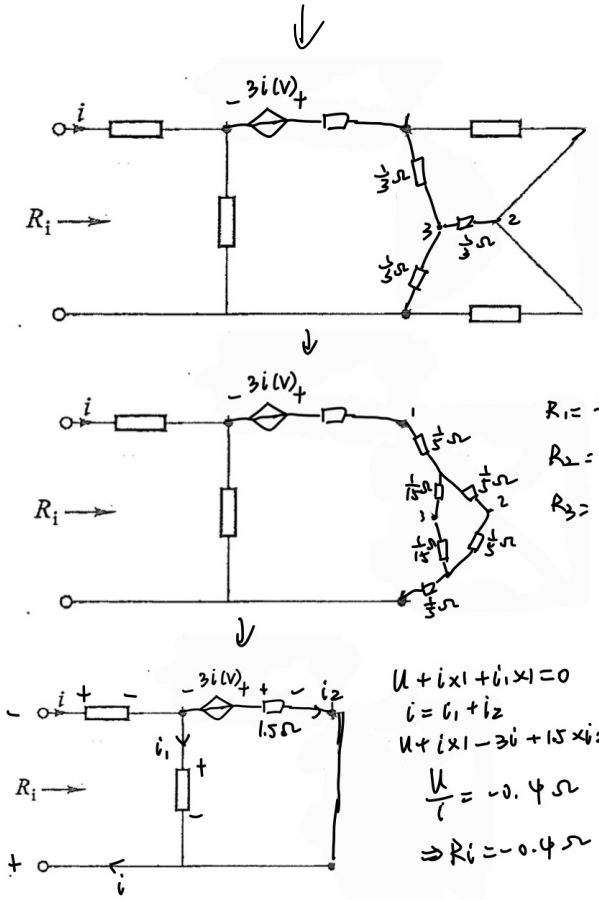


题 2-15 图

2-17 题 2-17 图所示电路中全部电阻均为 1 Ω，求输入电阻 R_i。



题 2-17 图



$U + i \times 1 + i_1 \times 1 = 0$

$i = i_1 + i_2$

$U + i \times 1 - 3i + 1.5 \times i_2 = 0$

$\frac{U}{i} = -0.4\Omega$

$\Rightarrow R_i = -0.4\Omega$